

Batch Code: ANP_D1838

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LAB-04

Question:

1. Write a java program to print alphabets A-Z using while loop
2. Write a java program to print reverse A-Z using do. while loop
3. Write a java program to print number of uppercase, lowercase, digits, spaces.

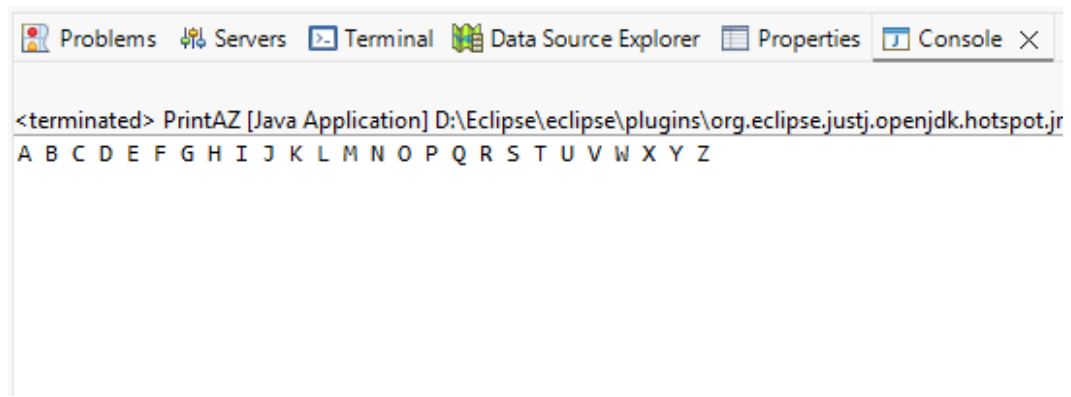
Input - WelCoME To HaPPy CheNNAI 600035 ToP Metro City

1. Print A–Z using while loop

Program:

```
public class PrintAZ {  
  
    public static void main(String[] args) {  
  
        char ch = 'A';  
  
        while (ch <= 'Z') {  
  
            System.out.print(ch + " ");  
  
            ch++;  
  
        }  
  
    }  
  
}
```

Output:



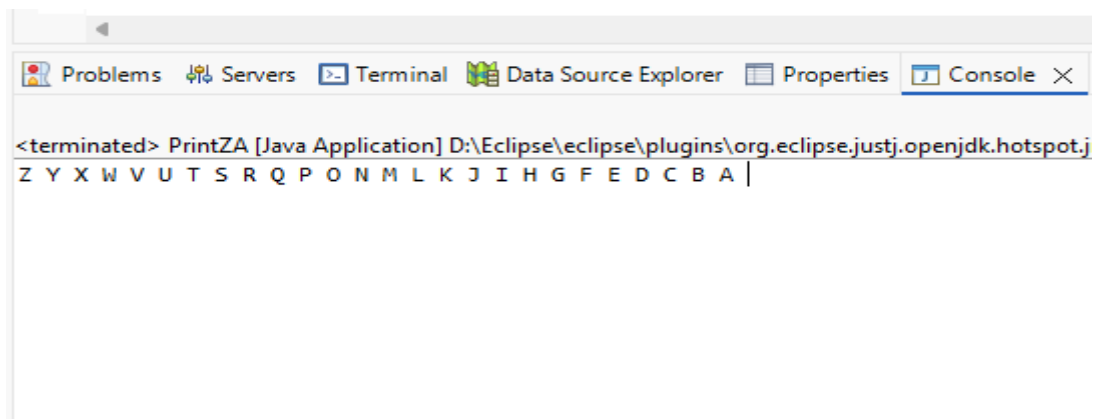
The screenshot shows the Eclipse IDE's console window. The title bar includes tabs for Problems, Servers, Terminal, Data Source Explorer, Properties, and Console. The console text reads:
<terminated> PrintAZ [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jr
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

2. Print Z–A using do-while loop

Program:

```
public class PrintZA {  
    public static void main(String[] args) {  
        char ch = 'Z';  
        do {  
            System.out.print(ch + " ");  
            ch--;  
        } while (ch >= 'A');  
    }  
}
```

Output:



The screenshot shows the Eclipse IDE's console window. The title bar includes tabs for Problems, Servers, Terminal, Data Source Explorer, Properties, and Console. The console text reads:
<terminated> PrintZA [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.j
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A |

3. Count Uppercase, Lowercase, Digits, Spaces

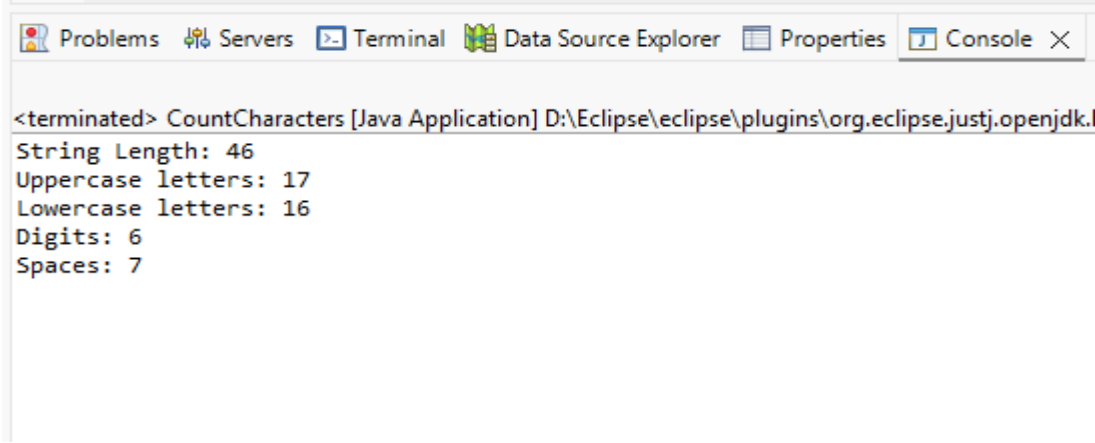
Input: "WelCoME To HaPPy CheNNAI 600035 ToP Metro City"

Program:

```
public class CountCharacters {
    public static void main(String[] args) {
        String str = "WelCoME To HaPPy CheNNAI 600035 ToP Metro City";
        int upper = 0, lower = 0, digits = 0, spaces = 0;
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            if (ch >= 'A' && ch <= 'Z')
            {
                upper++;
            }
            else if (ch >= 'a' && ch <= 'z')
            {
                lower++;
            }
            else if (ch >= '0' && ch <= '9')
            {
                digits++;
            }
            else if (ch == ' ')
            {
                spaces++;
            }
        }
        System.out.println("String Length: " + str.length());
    }
}
```

```
        System.out.println("Uppercase letters: " + upper);
        System.out.println("Lowercase letters: " + lower);
        System.out.println("Digits: " + digits);
        System.out.println("Spaces: " + spaces);
    }
}
```

Output:



The screenshot shows the Eclipse IDE's Console window. The title bar includes tabs for Problems, Servers, Terminal, Data Source Explorer, Properties, and Console. The console output is as follows:

```
<terminated> CountCharacters [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.l
String Length: 46
Uppercase letters: 17
Lowercase letters: 16
Digits: 6
Spaces: 7
```